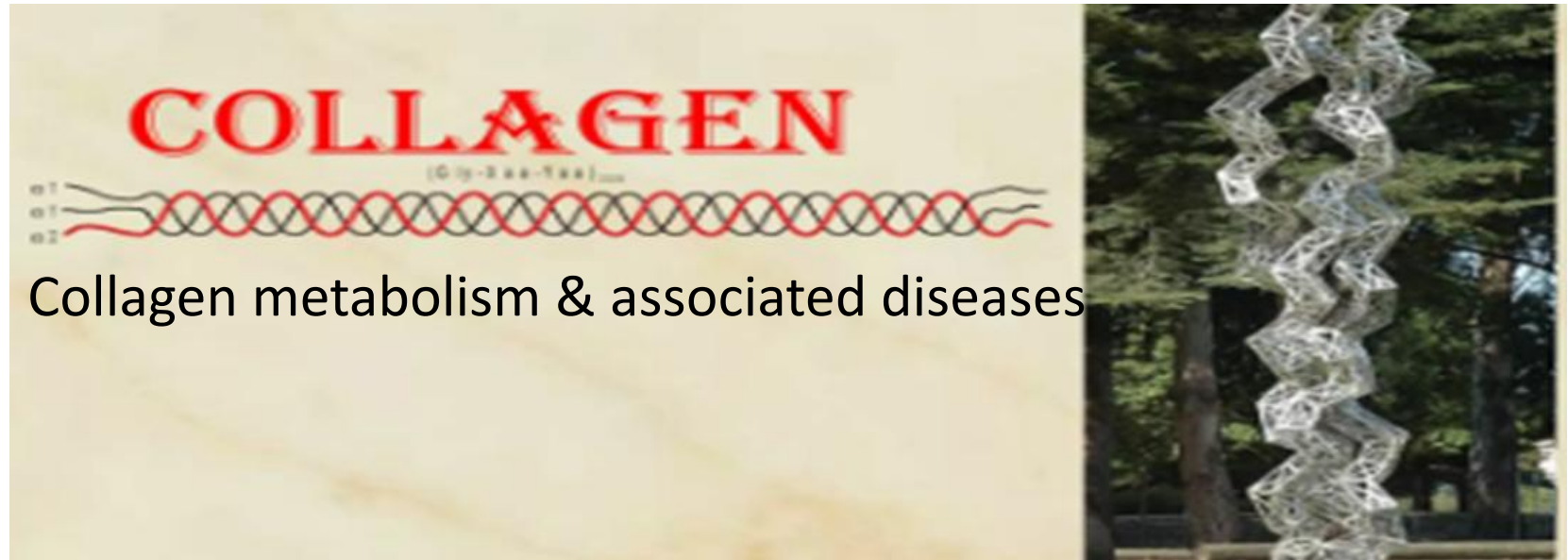
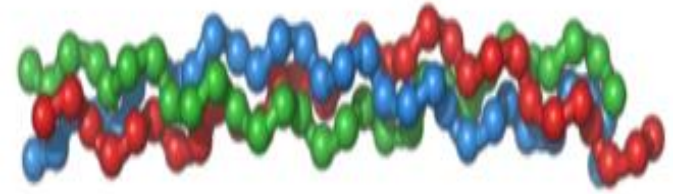


Collagen



Collagen metabolism & associated diseases

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Introduction Collagen

- The word collagen is derived from Greek word which means to produce glue /bond
- Collagen alone makes up 33% of the protein composition of the whole body
- it is the major component of the body's various connective tissues
- Collagen synthesized in two phase
- Collagen in ECM synthesized & secreted by mainly by connective tissue cells, fibroblast
- Collagen is the main structural protein in the extracellular matrix & characterized by its unique triple helix

Introduction Collagen

- Extracellular matrix is composed of three main proteins
 - namely, collagen, non-collagen and proteoglycan
- Collagen is a primary a building block */a structural components of various c.tissues &which is present*
 - in skin, muscles, tendons, ligaments, cartilages, and organic matrix of bones
- Mainly it forms a structural layer under the skin, and
 - under the skin cells, helps to renew and repair themselves

Introduction Collagen...

- The fibrous component of connective tissue are collagen
- w/c provide high tensile strength, resistance to structure of various tissues of the body and cannot be stretched
- The other elastin, unlike collagen it can be readily stretched
- It should be noted that the term fibroblast is often used interchangeably with fibrocyte
- Technically, however, a fibrocyte is the inactive precursor of a fibroblast

Introduction Collagen...

- With age, the structural support provided by collagen becomes weak or change over the years:
 - ✓ collagen fibers are synthesized in smaller quantities
 - ✓ existing collagen fibers undergo progressive degradation
- It can lead to:
 - a loss of elasticity in the skin and other connective tissues
 - lines, wrinkles, and looseness in the skin & joint stiffness
- When collagen levels are high, the skin is soft, smooth, and firm

Structure Collagen ...

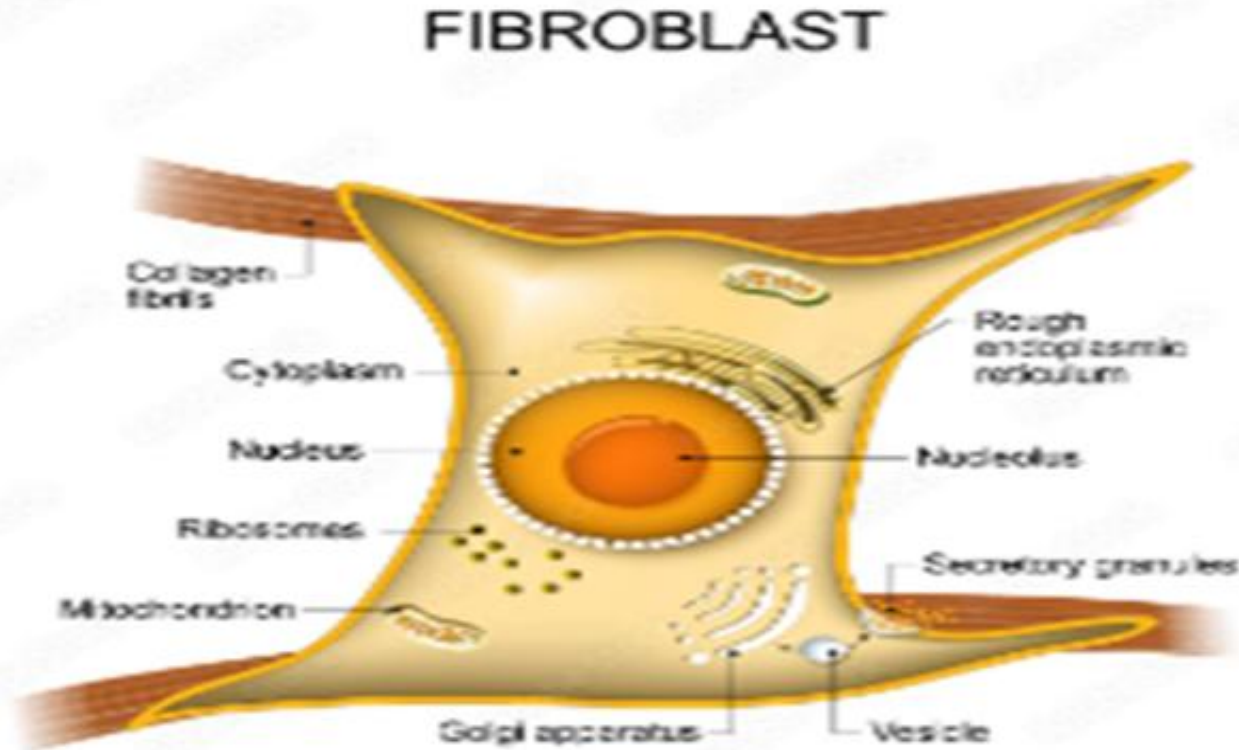
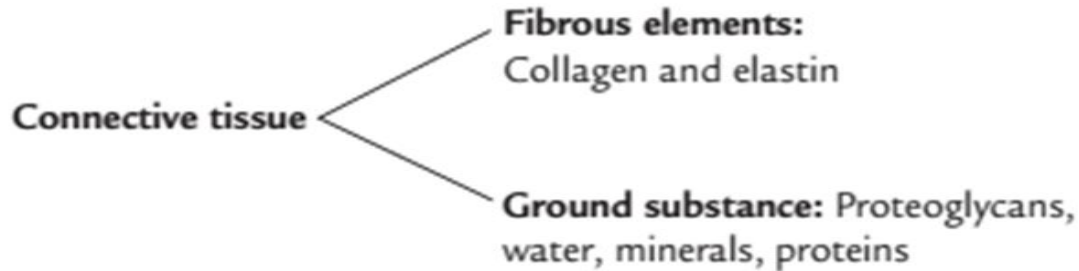


Fig 1 .The process of collagen synthesis occurs mainly in the cells of fibroblasts. Collagen is the most abundant type of protein in the body and a key component of the skin. It plays a key role in many bodily functions, including wound healing.

Composition Collagen...

- Connective tissue comprises of



ECM composed of fibers elements & ground substance

- Collagen is present in all human tissues and organs where it serves as **the major structural component**

Ground substance

- in which various **fibrous elements** of connective tissue are embedded
- The ground substance acts as a cushion or a lubricant
- Proteoglycans are hybrid molecules of proteins and polysaccharides(GAGs)
 - **w/c form the ground substance**

Composition Collagen...

- **Madras** described the structure of Collagen as having,
 - 3- α -polypeptide chains which are
 - **coiled around each other to form a right-handed triple helix** structure which is widely accepted even today
- The three polypeptide chains could be identical or different in collagen
- In most case It consist,
 - two identical $\alpha 1$ & the third $\alpha 2$ chain w/c are slightly d/n in their composition

Structure and composition Collagen...

- What make Collagens distinct is its unusual amino acid sequence, and by high proportion of certain amino acids: 33% glycine and 10% proline & the rest by other sometimes they are hydroxylated amino acids like lysine
- A common repeating sequence in Collagens
Glycine- poline-x or
Glycine-x –hydroxypoline where X represent any a.acids
- This Specific amino acid sequence made collagens distinct , contribute to strength & stability

Composition Collage...

- Each of these amino acids plays a specific role in folding of the polypeptide chain, as follows:
 1. Proline residues introduce sharp bends in the α -chains (because R group of proline is part of a cyclic ring),
 - which helps **tight wrapping** of these chains around one another
 - ✓ Such tight wrapping enhances strength of the triple
 2. Glycine residues play an indirect role in permitting extremely **tight interwinding** of the α -chains
 - ✓ The winding is so compact that there is little space available towards the interior
- it is the smallest amino acid, i.e can be accommodated at that place

composition Collage...

- Importance of glycine in collagen is, substitution of this amino acid by a bulkier one results in
 - **marked reduction in the strength of Collagen**

3. Hydroxyproline and Hydroxylysine are modified amino acids in collagen,

✓ each containing an additional hydroxyl group

- Each hydroxyl group in turn can participate in the formation of an additional hydrogen bond

Biochemical Functions Collagen

- Roles of collagen include:
 - Forms connective tissue framework (bone, skin, tendons)
 - Provides tensile strength
 - Structural Support: Collagen fibers provide the SCaffolding for tissues, maintaining their integrity and function.\
 - Cell Adhesion: This protein facilitates the attachment of cells to the extracellular matrix
 - Mechanical Resilience: Collagen resists tensile forces well, which is essential for organs that
 - **experience stretching or pulling**
 - Required for tissue repair and wound healing

Biosynthesis of Collagen

- Biosynthesis of Collagen in two phase
- Intracellular & extracellular phase
- Collagen is not synthesized from fibroblasts alone but by various other cells such as
 - **chondroblasts, osteoblasts, muscle cells, epithelial cells, endothelial cells and Schwann cells**
- Although these cells secrete collagen in the same manner as that of fibroblasts, the types of collagen they secrete vary
- fibroblasts are specialized cells with the main function of synthesizing collagen and stroma

Biosynthesis of Collagen

Cellular Level

Intracellular of fibroblasts cell

- Transcription of mRNA from portion of DNA start in nucleus
- Genes for pro- α 1 and pro- α 2 peptides chains are transcribed

Translation

- mRNA moves from the nucleus to the cytoplasm and interacts with ribosomes on the rough endoplasmic reticulum for translation
- Ribosomes read genetic code on mRNA as triplet code in the mRNA, for incorporating one specified amino acid
- After translation, it is referred to as pre-pro-polypeptide chain or leading to the formation of 3- α -polypeptide chains

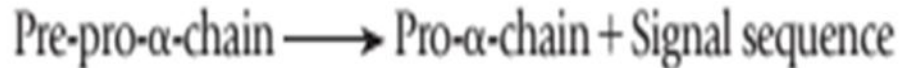
Biosynthesis of Collagen...

Post-translational modification

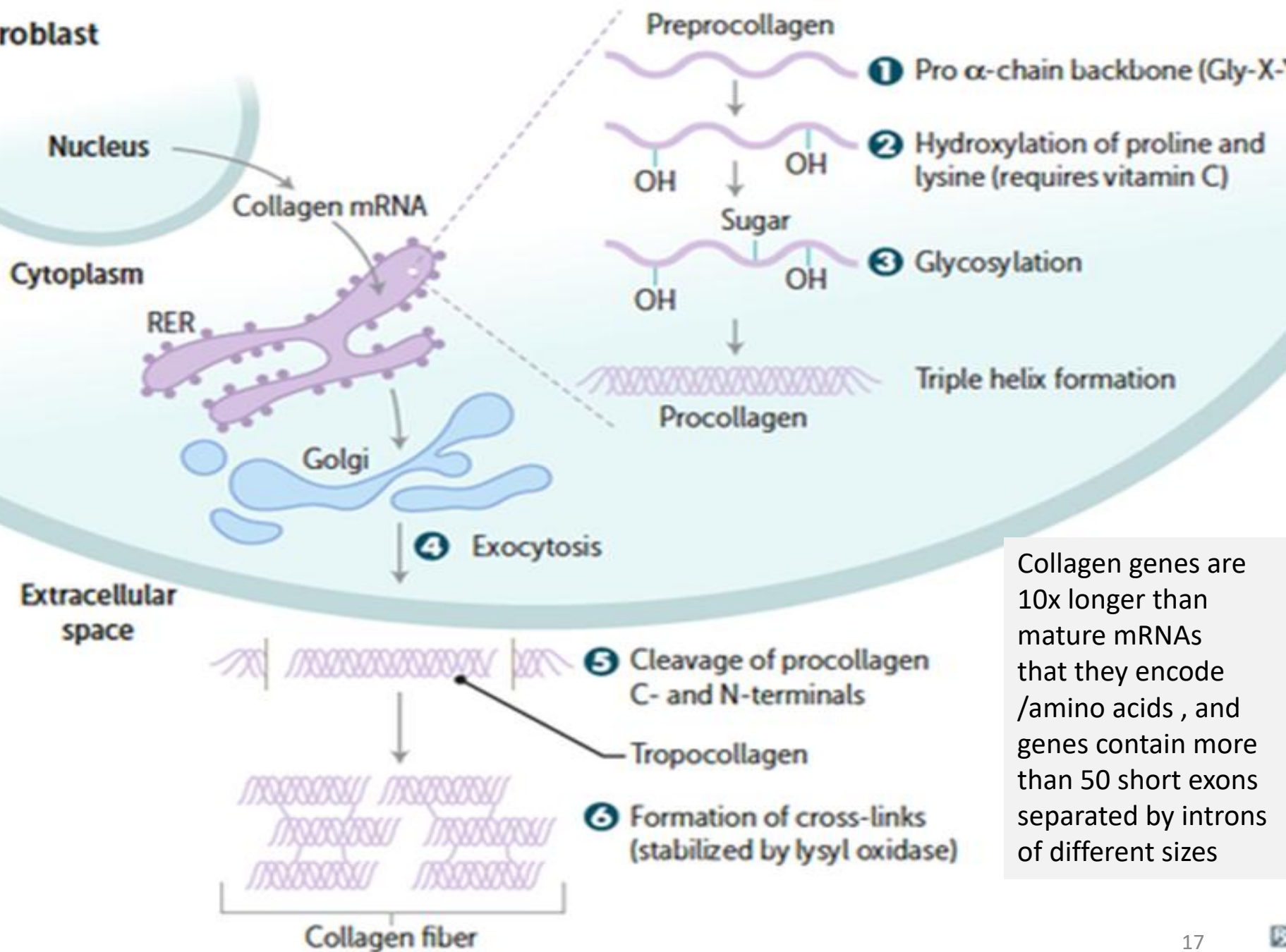
Once pre-pro-polypeptide enter the r.ER, **three major modifications** are made to the pre-pro-polypeptide for it to become pro-collagen

1. Nascent chain initially synthesized by translation, is known as pre-pro- α -chain

It contains a sequence of 15–30 hydrophobic amino acids/ signal peptide at its N-terminal end is removal from chain , to form pro- α -chain



Fibroblast



Collagen genes are 10x longer than mature mRNAs that they encode /amino acids , and genes contain more than 50 short exons separated by introns of different sizes

Collagen synthesis

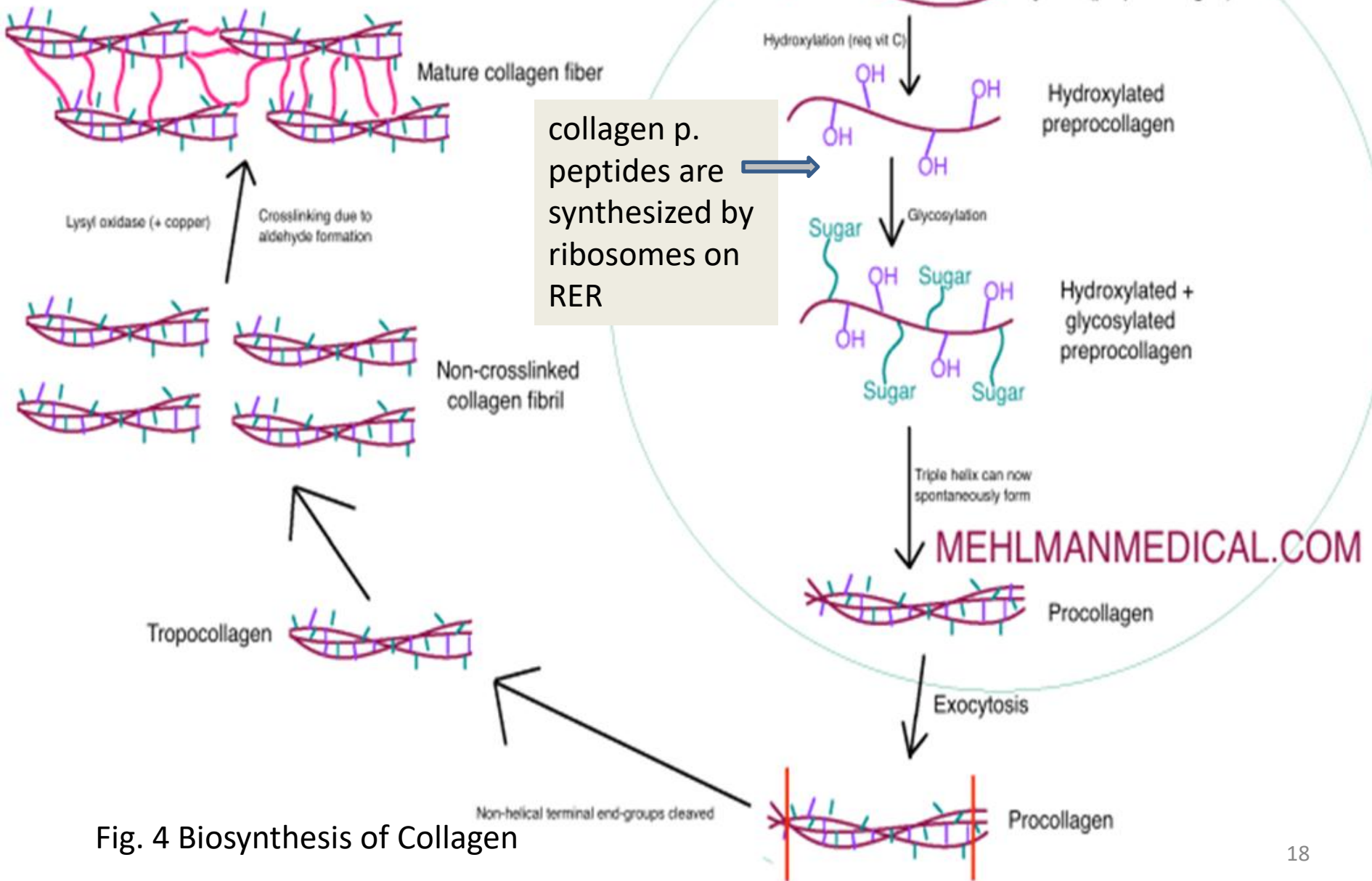


Fig. 4 Biosynthesis of Collagen

Biosynthesis of Collagen...

2. Hydroxylation

- The lysine and proline residues get additional hydroxyl groups added to them via hydroxylase enzymes
- These enzymes have an Fe^{2+} ion at their active site and require ascorbic acid (vitamin C) for activity
- The ascorbic acid acts as an anti-oxidant, keeping the Fe^{2+} ion in its reduced state
 - Both enzymes are dioxygenases, using molecular oxygen
- α -Ketoglutarate, the TCA cycle intermediate is a co-reductant, which is oxidized to succinate during the reaction (Fig.3)

Biosynthesis of Collagen...

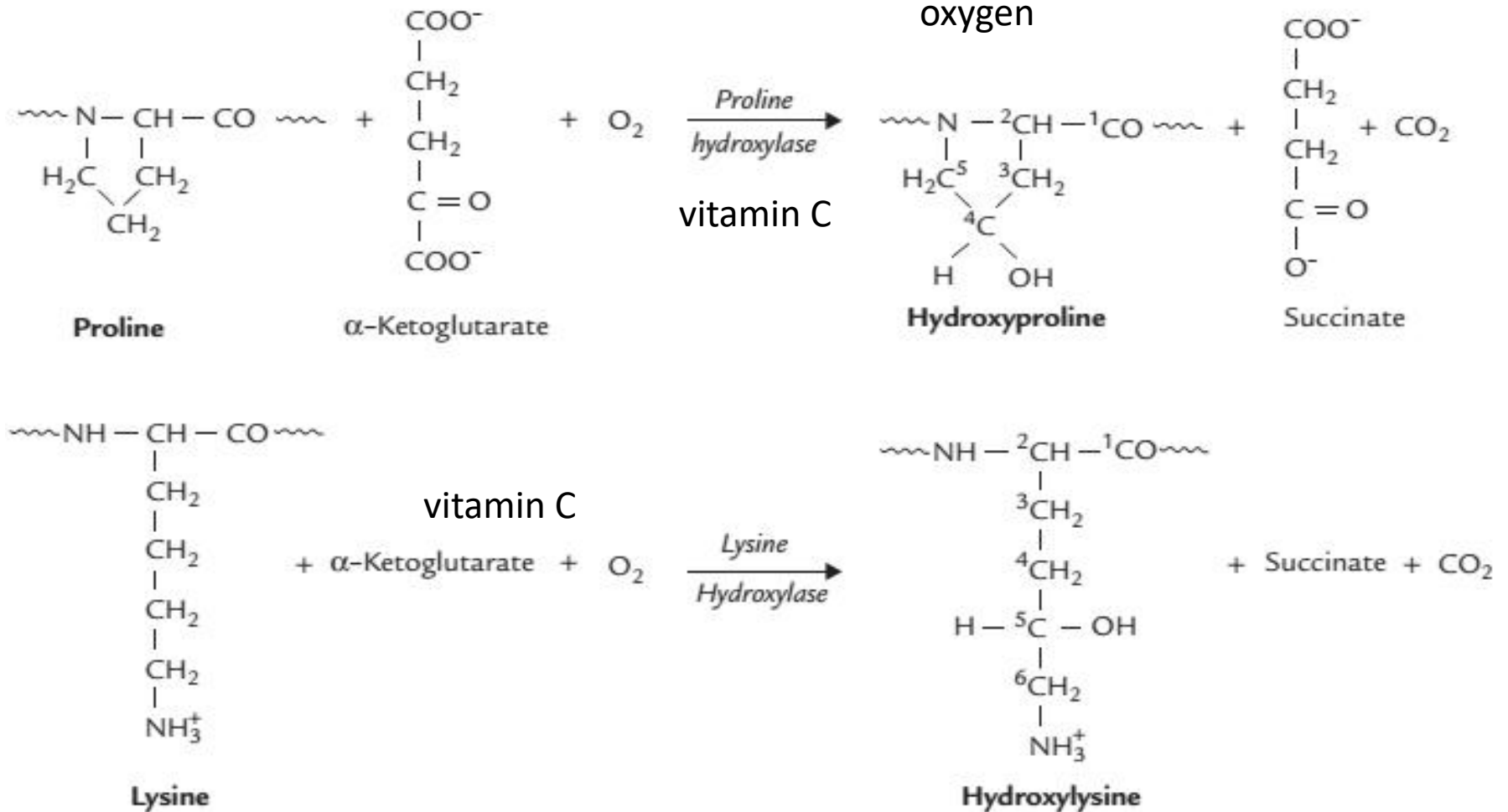


Fig. 5. Hydroxylated amino acids in collagen, synthesized post-translationally from proline and lysine residues in the polypeptide. Contributes to secondary structure (H-bonding)

Biosynthesis of Collagen...

- Because the hydroxylated amino acids are important for stabilizing the structure of collagen through hydrogen bond formation
- in vitamin C deficiency decreased synthesis of the hydroxylated amino acids results **Osteogenesis Imperfecta (OI)**, in weakening of the collagen fibers
 - defective hydrogen bonding
- This leads to the fragile blood vessels, incomplete bone formation and poor wound healing
 - the characteristic of the disease scurvy

Biosynthesis of Collagen...

Glycosylation

- Collagen is unique in having ***covalently*** attached
 - monosaccharides (galactose) or
 - disaccharides (glucose-galactose)
 - linked to the 5-hydroxyl of lysine
- The attachment are in endoplasmic reticulum by Mn containing UDP-glycosyltransferases
- Like hydroxylation, glycosylation occurs concomitantly with p. chain synthesis
- Glycosylation Contributes , *tertiary structure*
- More disulfide bonds form between polypeptides

Biosynthesis of Collagen...

Procollagen Triple Helix Formation

- **Three pro-peptide chains** assemble together to form the triple helix , **occurs in the R.endoplasmic reticulum**
- the three hydroxylated and glycosylated pro- α -chains assemble by;
 - **twisting into a triple helix by zipper-like folding**
- Now the pro-collagen molecule is ready to move to the Golgi apparatus for final modifications, glycosylation and assembled into
 - **secretory vesicles** to enter the extracellular space from fibroblast

Biosynthesis of Collagen...

Extracellular phase of synthesis of collagen

- This phase starts after Procollagen triple helix is exocytosed to extracellular space from fibroblast
- Initially when the collagen polypeptides are synthesized,
- they have additional amino acid residues (100–300) on both their N and C termini,
 - w/c are called extension peptides
- The extension peptide bonds are held together by disulphide bond formation
- These are removed off Procollagen triple helix to form tropocollagen, mature collagen fiber (Fig.6)

Biosynthesis of Collagen...

- The structure of collagen also has non- collagenous domains which are
 - important for the structural stability of the molecule
- These domains are called -C and -N terminals of the molecule
- The -C terminal is involved in initiation of the polypeptide chain formation while
- the -N terminal is associated with the regulation of the fibril diameter

Biosynthesis of Collagen...

- The extension peptides also prevent the premature aggregation of
 - **the pro collagen triple helices within the cell**
- the extension peptides are removed by the action of extracellular peptidases called Procollagen peptidase
- It is cleaved off N- and C-terminal regions, thereby
 - **decreasing the procollagen's size**
 - **decreasing solubility in water**
 - **If these regions are not cleaved off, ↑ H₂O solubility weakens collagen**

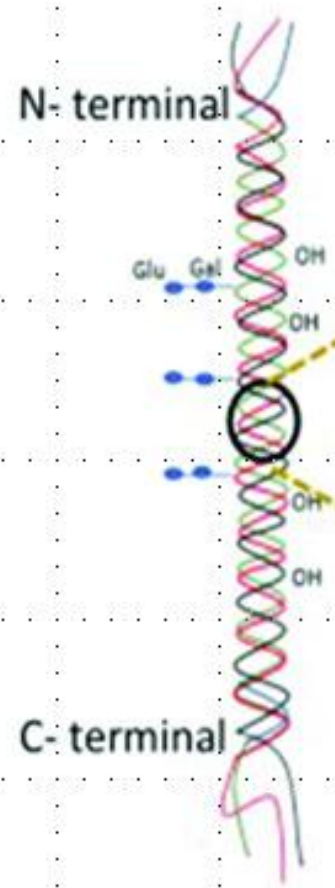
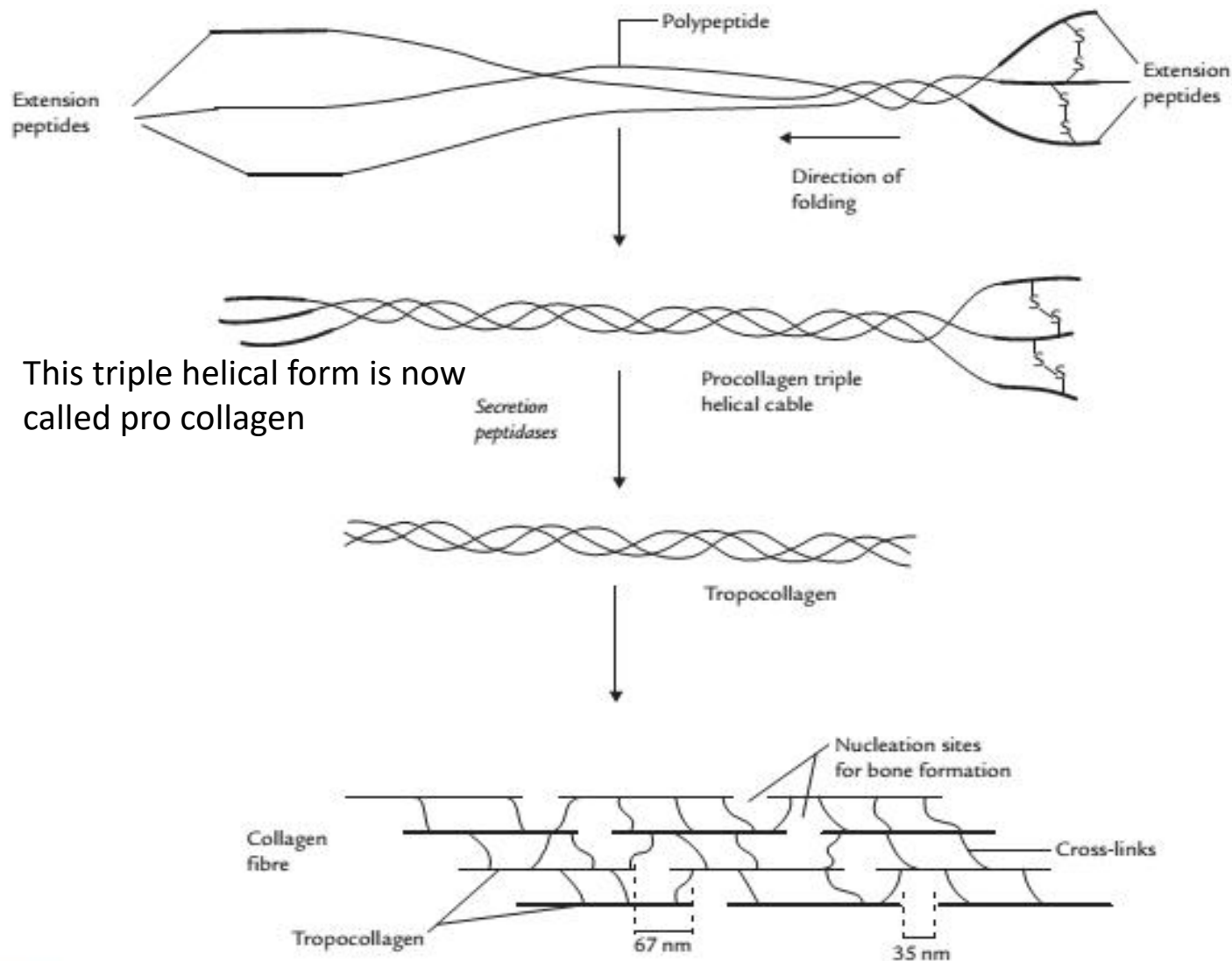


Fig. 6. Formation tropocollagen form procollagen . The tropocollagen molecules aggregate to form collagen fibrils, which cross link extensively to form collagen fibres.

Biosynthesis of Collagen...

- Tropocollagen arranges to form aggregate into closely packed collagen fibrils
- **Collagen fibrils** are then cross linked to form collagen fibers
- Ehlers-Danlos syndrome (usually ↓ collagen III)
- It cause due to cross linking problem

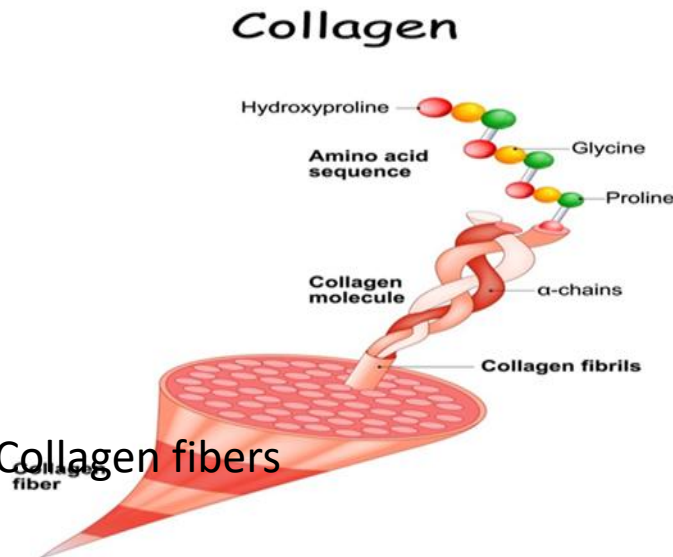


Fig .7: Collagen fibers

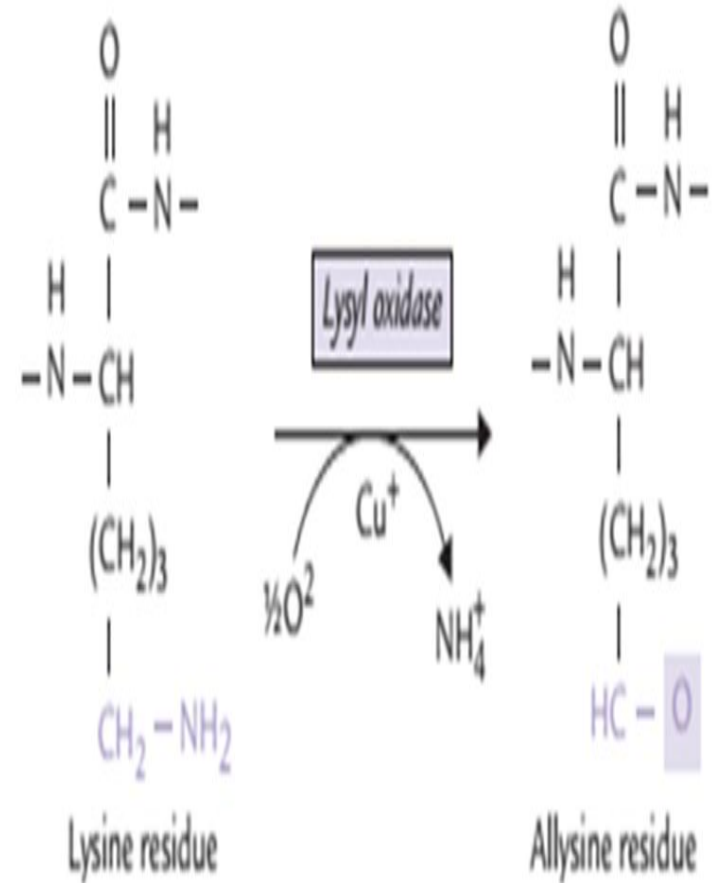


Fig. 7.The lysyl oxidase reaction

Biosynthesis of Collagen...

- b/n collagen fibrils covalently bond are formed by oxidative deamination of the lysine and 5-hydroxylysine residues by lysyloxidase a copper-dependent enzyme
- Lysyl oxidase creates aldehydes from oxidative deamination of the above amino acid (reduced molecule)
- Menkes syndrome leads to,
 - ↓ activity of lysyl oxidase because of copper deficiency

Types of Collagen and Their Functions

- Collagen is protein; it actually exists in several different forms, & found in different tissues, each for specific functions.
- The most common types include:
- Type I: This is the most prevalent type, found in bones, skin, tendons and ligaments. It offers high structural integrity.
- Type II: Mainly present in cartilage, it is vital for joint health, providing cushioning and support.
- Type III: Often found alongside type I collagen, it contributes to skin elasticity and is significant in blood vessels.
- Type IV: This type forms the basal lamina, a layer that serves as a foundation for various tissues.
- Understanding these various Collagen type /forms gives us how different collagen pathologies can manifest

Degradation of Collagen

- Native collagen is resistant to most tissues proteinases & digestive enzymes
- In normal tissues, along with the synthesis of collagen, its degradation followed by replacement with new fibers,
- Collagen tissue undergoes constant & slow turnover
 - half-life being up to week -months
- Normal degradation of collagen occurs during
 - tissue repair , during normal growth and development, and is initiated via the collagenases enzymes
 - It is constant remodeling of the connective tissue
- The imbalance b/n synthesis and degradation of collagens lead to various pathological condition of connecting tissues

Degradation of Collagen...

- Collagen is cleaved in two locations, intracellular and extracellular
- The extracellular collagenases contains zinc, initiate degradation by splitting
 - a single Gly-leucine or isoleucine bond in each of the three tropocollagen strands
- The collagen fragments produced are water soluble
 - easily denatured and become susceptible to attack by several intracellular lysosomal proteases (e.g. cathepsins)



Classification of Collagen Disorders

- Based on their origin and clinical manifestations, they are commonly classified into two categories:
- Genetic /heritable disorders and acquired conditions
- Both disrupt the normal structure or function of collagen in body.

1. Genetic Defects of Collagen Metabolism

Heritable Disorders: These are due to mutations in the genes responsible for collagen production.

Genetic Defects of Collagen Metabolism

- Inherited abnormalities in collagen metabolism include;
 - abnormal collagen genes
 - abnormal post translational modification of collagen, or
 - deficiency of cofactors needed by the enzymes that carry out the post-translational modifications
- Defective collagen synthesized in these diseases is unable to provide the required tensile strength
- Examples include Ehlers-Danlos Syndrome, Osteogenesis Imperfecta

Genetic Defects of Collagen Metabolism...

Osteogenesis Imperfecta (OI)

- It is a group of genetic disorders /a mutation in which biosynthesis of type I collagen is defective
 - Fragile bones, weak bones and irregular connective tissue & blue sclera
 - commonly known as brittle bone disease
- in vitamin C deficiency decreased synthesis of the hydroxylated amino acids
 - results Osteogenesis Imperfecta (OI)

Genetic Defects of Collagen ...

Epidermolysis Bullosa

- It is a heritable disease in which glycosylation of collagen is impaired and the skin characteristically blisters

Ehlers-Danlos syndrome (EDS)

- The condition can arise from mutations in genes responsible for collagen production
- Collagen synthesis/cross-linking defects
- Hyperextensible skin, joint laxity
- It is primarily characterized by a defective collagen synthesis process, leading to an abnormal structure of connective tissues.

Genetic Defects of Collagen Metabolism...

Dermatosparaxis

- There is a hereditary impairment in the removal of the terminal extension propeptides by Procollagen peptidase
- Because fibril formation can occur only when these extension peptides have been removed, otherwise
- there is impairment in fibril formation and hence, reduced tensile strength of collagen

Classification of Collagen Disorders...

2. Acquired conditions: Unlike hereditary disorders, these may arise due to environmental factors, inflammation, or metabolic changes

- Conditions like rheumatoid arthritis can lead to collagen damage, affecting its function in joints and tissues
- Osteoporosis – Not inherited, brought with age, associated with reduced levels of collagen in the skin and bones

Disorders of Collagen Degradation

- Most collagen disorders are characterized by abnormal activities of collagen-degrading enzymes
- Their activity is low in normal human tissues, but in some *pathological states* characterized by,
- increased collagen degradation, such as;
 - Paget's disease
 - Osteoarthritis
 - rheumatoid arthritis, and
 - tumor metastasis, higher activity of these enzymes occur

Effect of Aging on Collagen Metabolism

- The meat of young animal tastes better, soft and tender, whereas that of old animals is tough and unpalatable
- The reason for this is, there is age related change occur on Collagen that is
 - **the collagen fibrils of old animals have more covalent cross links than those of the young**
- Also the amount of collagen, relative to the proteins of parenchymal cells, increases with age

Summary

- During wound healing, the fibroblasts migrate to the site of injury and produce abundant collagen
- Collagen synthesis is also stimulated by tissue injury
- Scar tissue consist mostly of collagen
- The death of **parenchymal tissue** is followed by deposition of a collagen-rich extracellular matrix

Summary...

Role of Collagen in the Body

- Collagen is a substance that the body produces naturally.
- Collagen comprises approximately 30% of the protein content in the human body, making it the most abundant protein present.
- It serves as the primary building block for a host of tissues, offering both tensile strength and flexibility.
- Think of it as the sturdy scaffolding that holds up a building. Whenever this scaffolding/ ladder becomes weak or compromised, structural failures ensue, leading to a plethora of health issues.

Summary ...

- Collagen metabolism is affected in a number of other diseases, e.g. scurvy and lathyrism.
- Copper deficiency also leads to enzyme inhibition resulting in failure of collagens to acquire the tensile strength they need to function.
- Functions of collagen include keeping the skin smooth and elastic, keeping the joints flexible, and enabling wounds to heal
- Eating foods that contain vitamin C and antioxidants, avoiding smoking, limiting caffeine intake, and protecting the skin from sunlight may all help preserve collagen or boost its production

NB: Read more on Intracellular & extracellular degradation of collagen

Thanks !!